SNOWFLAKE DATA ENGINEERING PROJECT

# Healthcare Intelligence Hub

Patient, Cost & Care Analytics

## Phase 7 — Project Documentation (Final Report)

Institutional template-based phase document


| | |
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Prepared from the verified project source (query.sql, Health/streamlit_app.py, Health/snowflake.yml, six CSV datasets). Bracketed fields are placeholders for information not supplied.

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# Phase 7 — Project Documentation (Final Report)

**Project:** Healthcare Intelligence Hub: Patient, Cost & Care Analytics **Template covered:** Project Report Format (Final Report Template). Sections 1–11 below follow the template order exactly. Each section summarises its phase and cross-references the detailed phase document.

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**Abstract.** This project builds a Snowflake data-engineering pipeline and a Streamlit-in-Snowflake dashboard. They address a clear problem: hospital administrators and finance teams need a consolidated, queryable view of patient visits (cost, length of stay (LOS), satisfaction and readmissions) that today sits fragmented across raw operational records. The project answers which facilities and doctors perform well, which diagnoses drive the longest and most expensive stays, and where patient satisfaction is at risk. Six CSV sources (14,059 rows) are loaded into `HEALTH_DB`, cleaned in `STAGING`, modelled as a visit-grain star schema in `CURATED`, and summarised by ten `MART` views. Fifty-five automated checks were executed against a local replica of the pipeline: 50 pass, and 5 expose genuine data or logic issues (two `INITCAP` label changes, a doctor-grouping grain error and two source-data anomalies). Snowflake-side and live-UI tests are scripted but not yet executed.

**Keywords:** Snowflake, data warehouse, ELT, star schema, Streamlit, healthcare analytics, readmission, data quality.

## 1. INTRODUCTION

### 1.1 Project Overview

The Healthcare Intelligence Hub is an end-to-end analytics solution built on Snowflake. Its components, all verified in the repository:

- `query.sql`: creates database `HEALTH_DB` with schemas `RAW`, `STAGING`, `CURATED` and `MART`; a CSV file format and an internal stage; 6 raw tables, 6 staging tables, 5 dimensions, 1 fact table and 10 views; and validation queries.
- `Health/streamlit_app.py`: a single-page dashboard with 6 KPI cards, 8 charts, 2 tables and 5 filters, reading `CURATED.FACT_PATIENT_VISITS` through the active Snowpark session.
- `Health/snowflake.yml` and `pyproject.toml`: deployment definition for Streamlit in Snowflake (warehouse `COMPUTE_WH`, Python 3.11).
- Six CSV data files covering January 2024 to August 2026.

### 1.2 Purpose

To replace fragmented raw operational records with a structured warehouse: a governed, re-runnable pipeline and a self-service dashboard. The aim is that visit volume, cost, LOS, satisfaction and readmission metrics are defined once and can be compared across facilities, doctors and diagnoses.

# 2. IDEATION PHASE

*(Full detail: Phase 1)*

## 2.1 Problem Statement

> Hospital administrators and finance teams need a consolidated, queryable view of patient visits — spanning cost, length of stay, satisfaction and readmissions — that today sits fragmented across raw operational records. Without a structured warehouse, it is difficult to answer basic questions such as which facilities and doctors are performing well, which diagnoses drive the longest and most expensive stays, and where patient satisfaction is at risk.

The customer view of this problem is captured in PS-1 to PS-3 (Phase 1 Table 1.3). The three questions are answered as follows (detail in Phase 2 Table 2.0):

| # | Question in the problem statement | How the solution answers it | Finding (full dataset, local replica) |
|---|---|---|---|
| Q1 | Which facilities and doctors are performing well? | `VW_FACILITY_ANALYTICS`, `VW_DOCTOR_ANALYTICS`; app facility chart and table, doctor leaderboard (visits, avg cost, satisfaction, readmissions) | Facility volume 958–1,039 visits; City General Hospital busiest (1,039). Highest doctor satisfaction: Dr. Neha Brown, D002 (4.02) |
| Q2 | Which diagnoses drive the longest and most expensive stays? | `VW_DIAGNOSIS_ANALYTICS` (avg cost, avg LOS); app filter Visit Type = Inpatient with diagnosis charts | Inpatient stays: longest Fracture (8.60 days avg), Anemia (8.32), Diabetes (8.07); most expensive Kidney Disease (14,816.82 avg), Hypertension (14,640.75), Asthma (14,549.67) |
| Q3 | Where is patient satisfaction at risk? | Avg Satisfaction KPI; satisfaction columns in facility and doctor tables and in `VW_DIAGNOSIS_ANALYTICS` | Lowest facility: Sunrise Hospital (3.78; 13.9% of visits score 1–2). Lowest diagnosis: Asthma (3.75). Lowest doctors: Dr. Rohan Singh, D060 (3.65), Dr. Rohan Gupta, D004 (3.69) |

## 2.2 Empathy Map Canvas

The combined administrator / finance persona *says* "I need one agreed number", *thinks* "are my joins right?", *does* monthly VLOOKUP consolidation, and *feels* overwhelmed and anxious about accuracy. The persona gains trust and speed from a single governed dataset (Phase 1 Table 1.4).

## 2.3 Brainstorming

Ten ideas were grouped into four themes. The three selected and implemented: (1) a layered Snowflake ingestion and cleaning pipeline; (2) a star schema with MART KPI views; (3) a Streamlit-in-Snowflake dashboard, chosen over Tableau to keep data and UI on one platform. Automation, RBAC and ML were deferred (Phase 1 Table 1.6).

# 3. REQUIREMENT ANALYSIS

*(Full detail: Phase 2)*

## 3.1 Customer Journey Map

Five stages: need → gather → prepare → present → follow-up. Each is mapped from today's manual pain points to the Hub's corresponding layer (Phase 2 Table 2.1).

## 3.2 Solution Requirement

Six functional epics: Data Access & Integration, Data Processing, Business Analysis, Business Insights, User Interaction, Data Update & Reporting. Six NFRs: usability, security, reliability, performance, availability, scalability (Phase 2 Tables 2.2–2.3). Data-quality requirements DQ-1 to DQ-6 were defined and then tested in Phase 6.

### 3.3 Data Flow Diagram

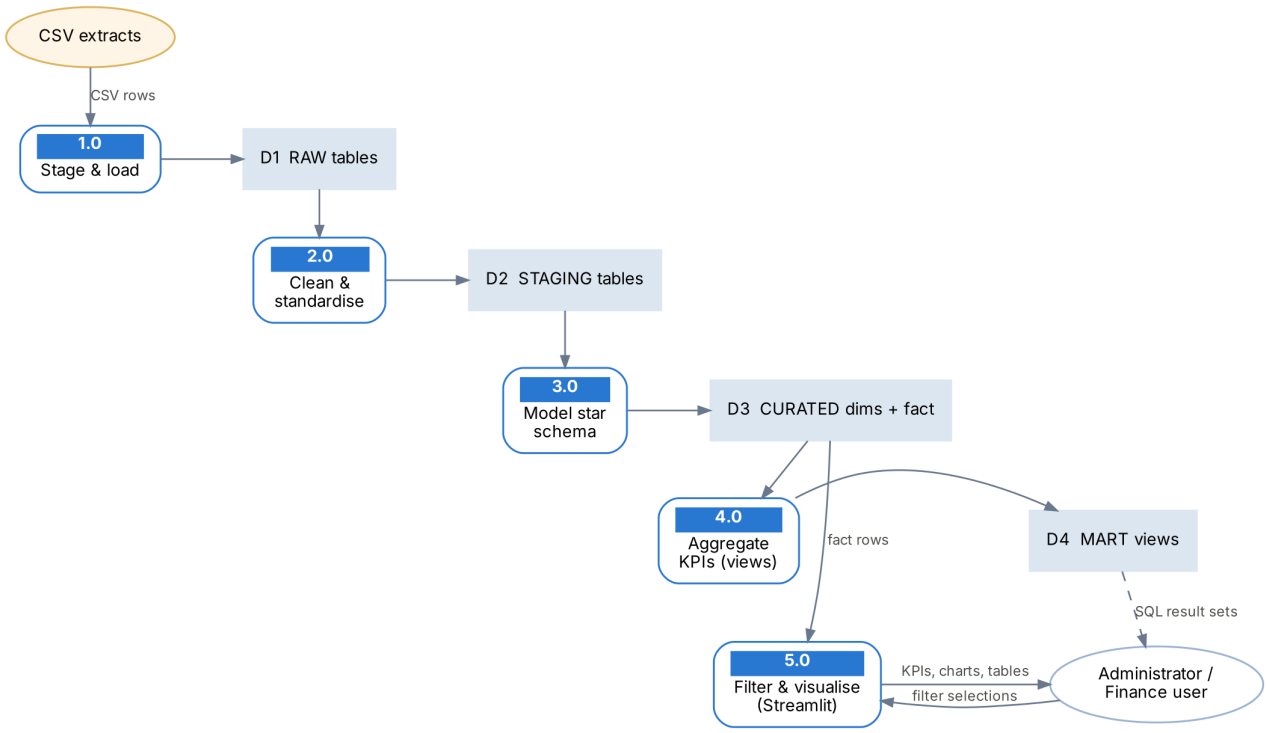



*Figure 7.1 — DFD Level 1 (see also Level 0 in Phase 2)*

### 3.4 Technology Stack

Snowflake SQL and Python 3.11; Snowflake `HEALTH_DB` with `COMPUTE_WH`; internal stage and file format for ingestion; Snowflake SQL (CTAS, views) and pandas for transformation; Altair and Streamlit for visualisation; Streamlit in Snowflake as the interface (Phase 2 Table 2.5).

## 4. PROJECT DESIGN

*(Full detail: Phase 3)*

### 4.1 Problem Solution Fit

The canvas links the customer segment (administrators and finance), constraints (time, skills, privacy), root cause (no integrated model or KPI definitions) and triggers (month-end reviews, cost or readmission spikes) to the solution and its online and offline channels (Phase 3 Table 3.1).

### 4.2 Proposed Solution

A Snowflake-native pipeline with a dashboard. Its novelty is a single governed platform with no credentials in the app. Its impact is earlier detection of high-readmission diagnoses and transparent out-of-pocket cost (≈27.7%). It scales by warehouse resizing, with SQL push-down proposed for large data (Phase 3 Table 3.2).

### 4.3 Solution Architecture

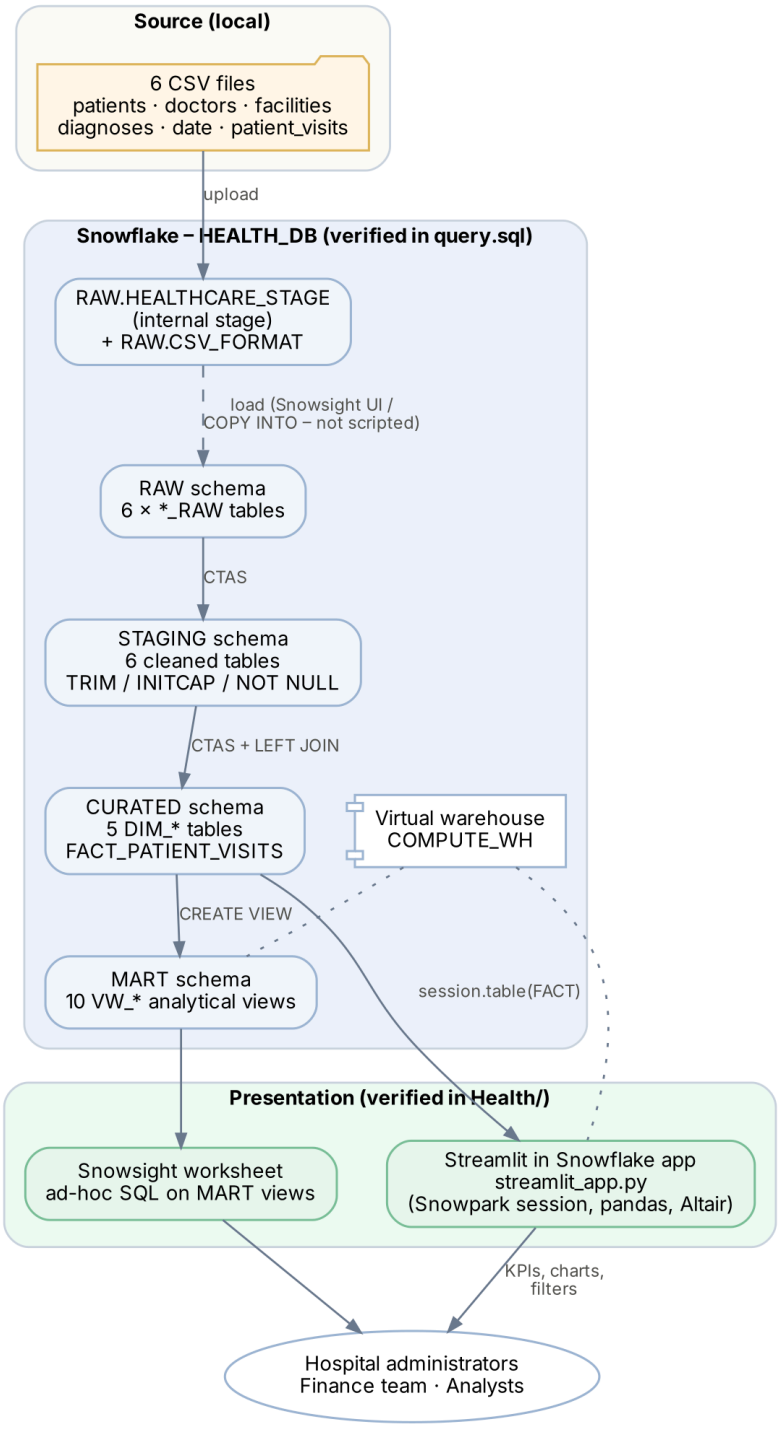



*Figure 7.2 — Solution architecture*

The star schema has `FACT_PATIENT_VISITS` (grain: one visit; 10,000 rows) and dimensions for patient (3,000), doctor (60), facility (10), diagnosis (15) and date (974). Keys are logical: no constraints are declared, and `DIM_DATE` is not yet joined (Phase 3 §3.6).

## 5. PROJECT PLANNING & SCHEDULING

*(Full detail: Phase 4)*

### 5.1 Project Planning

Four 6-day sprints of 15 story points each (60 in total; planned velocity 2.5 points per day). Sprint 1: setup and ingestion. Sprint 2: staging, curated model, MART views and validation. Sprint 3: KPI and chart stories. Sprint 4: filters, deployment, testing and documentation. Actual dates were not available in the project files and are left as placeholders.

## 6. FUNCTIONAL AND PERFORMANCE TESTING

*(Full detail: Phase 6 · TEST_CASES.md)*

### 6.1 Performance Testing

*Table 7.1 — Test summary*

| Suite | Cases | Executed | Pass | Fail | Not Executed |
|---|---|---|---|---|---|
| Local replica ( `tests/local_validation.py` ) | 55 | 55 | 50 | 5 | 0 |
| Snowflake ( `tests/snowflake_validation.sql` ) | 9 | 0 | — | — | 9 |
| Dashboard UI (manual) | 12 | 0 | — | — | 12 |

Defects (Phase 6 Table 6.6): D-01 doctor-view grain merges two doctors; D-02/D-03 `INITCAP` produces `Ent` and `Follow-Up` ; D-04 "Self Pay" visits show insurance coverage; D-05 admitted emergency visits with 0-day LOS; D-06 redundant view; D-07 unused `DIM_DATE` ; D-08 currency label; D-09 load not scripted. Local timings (≈0.3 s end to end) are not Snowflake measurements.

## 7. RESULTS

### 7.1 Output Screenshots

> The figures below were regenerated from the project CSVs with the same aggregation logic as the app and MART views. **They are not screenshots of the deployed Snowflake app.** Insert live screenshots at the marked placeholders (capture list: Phase 8 §8.8).

**[Screenshot placeholder — Streamlit app, unfiltered: header, sidebar filters and KPI row]**

*Table 7.2 — Headline KPI results (full dataset, local replica of VW_EXECUTIVE_KPI)*

| Total visits | Unique patients | Doctors | Facilities | Total cost | Insurance coverage | Patient payment | Avg satisfaction | Avg LOS | Readmissions |
|---|---|---|---|---|---|---|---|---|---|
| 10,000 | 2,889 | 60 | 10 | 82,818,797.00 | 59,918,913.84 | 22,899,883.16 | 3.83 | 0.97 | 1,203 (12.03%) |

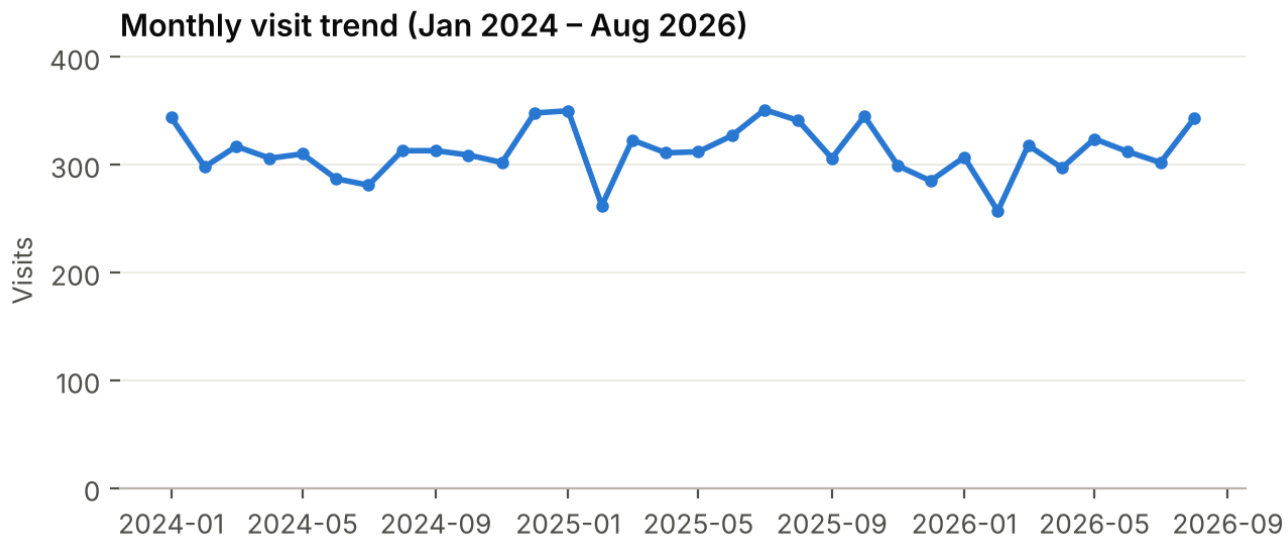



*Figure 7.3 — Monthly visit trend*

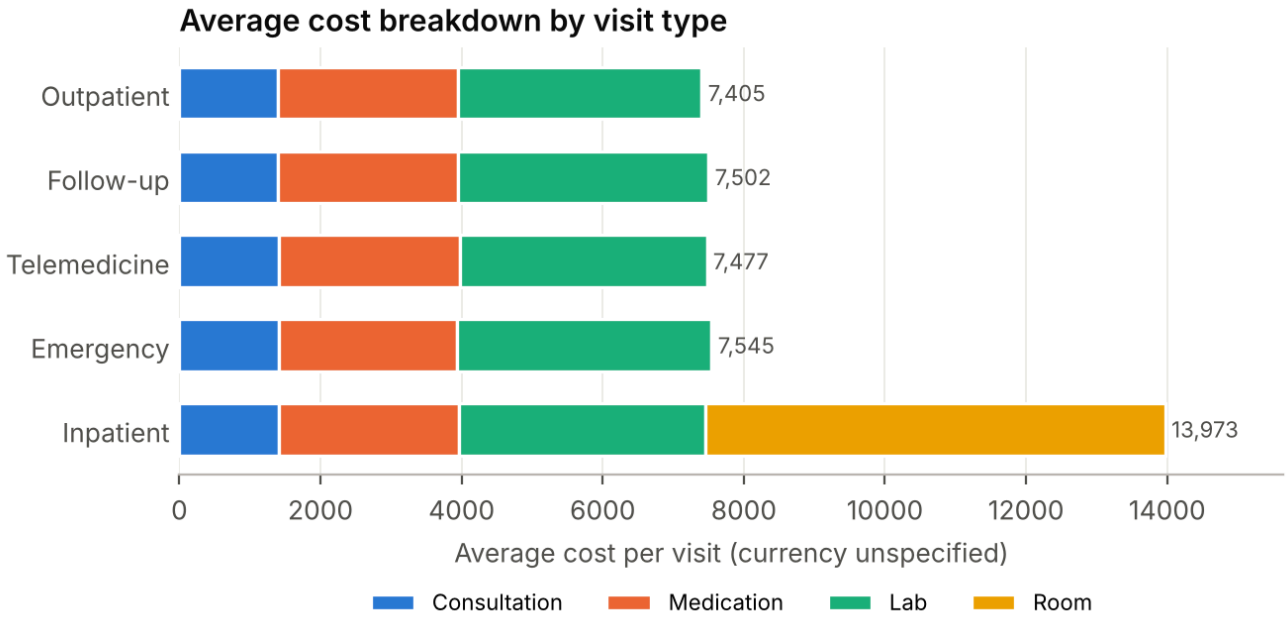



*Figure 7.4 — Average cost breakdown by visit type*

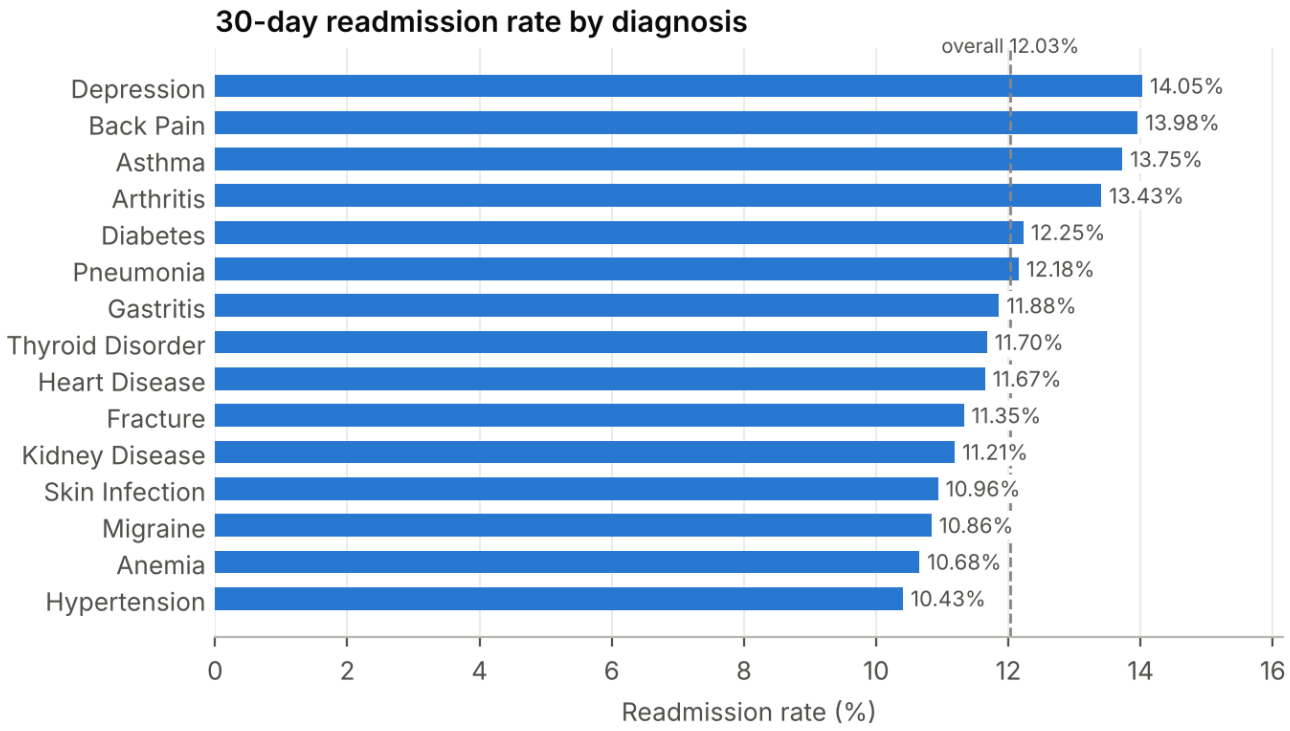



*Figure 7.5 — 30-day readmission rate by diagnosis*

**Discussion.** Inpatient visits (12.6% of volume) cost about 1.9× other types because only they incur room charges. Lab is the largest cost component (42.3%). Readmissions cluster in Depression, Back Pain, Asthma and Arthritis (13.4–14.1%, against 12.03% overall). Facility differences in volume, cost and satisfaction are small (cost per visit within about 4%), which is consistent with a synthetic dataset. Two source anomalies, insured "Self Pay" visits and zero-LOS admissions, mean that payer and LOS insights should be confirmed with the data owner before any operational use.

## 8. ADVANTAGES & DISADVANTAGES

| Advantages | Disadvantages / limitations |
|---|---|
| Whole stack in one platform: no data movement, no credentials in the app | Data load step not scripted; refresh is manual |
| Layered schemas make every metric traceable and rebuildable ( `CREATE OR REPLACE` ) | No RBAC or masking; patient names flow into the fact table and app |
| KPI logic centralised in 10 SQL views | App recomputes KPIs in pandas instead of using MART views, which duplicates logic |
| Interactive filters with caching; zero-row guards | Full fact table loaded into memory, which limits scale |
| Data verified clean: 0 NULL or duplicate keys, 0 orphans, costs reconcile | INITCAP label side effects; doctor-view grain bug; unused `DIM_DATE` ; redundant view |
| Low cost: X-Small warehouse suffices | Snowflake and UI tests not yet executed; synthetic data limits real-world conclusions |

## 9. CONCLUSION

The project meets its core objectives. It consolidates six healthcare sources into a validated star schema in Snowflake, defines visit, cost, LOS, satisfaction and readmission KPIs in reusable views, and presents them through a filterable Streamlit-in-Snowflake dashboard. Verification against a local replica confirmed complete, unique and

referentially intact data and correct KPI arithmetic. It also surfaced specific, fixable defects that the original validation queries did not catch. With the corrective actions in Phase 6 and the execution of the prepared Snowflake and UI tests, the solution would be ready for a pilot on real, governed data.

## 10. FUTURE SCOPE

See Phase 8 §8.11 for the roadmap. In summary: scripted `COPY INTO` with Snowpipe or Tasks for automated refresh; RBAC and masking policies; SQL push-down in the app; weekday and seasonality analysis via `DIM_DATE`; insurer and payer dashboards; SCD Type 2 dimensions; readmission-risk modelling with Snowpark ML; CI checks using the provided test scripts.

## 11. APPENDIX

**Source Code:** `query.sql` (complete SQL; key excerpts in Phase 5) · `Health/streamlit_app.py` (dashboard) · `Health/snowflake.yml`, `Health/pyproject.toml` (deployment) · `tests/local_validation.py`, `tests/snowflake_validation.sql` (tests added with this documentation). Data dictionary: `DATA_DICTIONARY.md`.

**Dataset Link:** the six CSVs are included in the repository root. Original dataset source: **[provide the source or "provided by institution"]**.

**GitHub & Project Demo Link:** GitHub: **[https://github.com/<username>/<repository>]** · Demo video: **[link]**

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